

CS 3120 Exam 2

This packet contains exam 1. This **cover sheet** is here to provide instructions, and to cover the questions until the quiz begins. **do not open this quiz packet** until your proctor instructs you to do so.

You will have 1 hour and 45 minutes to complete this exam. Each quiz is two pages (front and back of one sheet of paper) worth of questions. Make sure to **write your name and computing id at the top of each individual page**.

When you are finished, simply submit this packet at the front of the classroom.

This quiz is CLOSED text book, closed-notes, closed-calculator, closed-cell phone, closed-computer, closed-neighbor, etc. Questions are worth different amounts, so be sure to look over all the questions and plan your time accordingly. Please sign the honor pledge below.

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Module 3: Context-Free Grammars**Name** _____

1. [6 points] Answer the following True/False questions.

It is possible for a <i>context-free grammar</i> to generate only the empty string.	True	False
<i>PDA</i> s do not use <i>non-determinism</i>	True	False
A <i>PDA</i> 's stack can overflow, and in this case the machine always <i>rejects</i>	True	False
A <i>context-free grammar</i> exists for the language $0^*(10 \cup 01)^*0^*$	True	False
If a <i>PDA</i> has access to <i>two stacks</i> , then the language $a^n b^n c^n$ can be recognized by that <i>two-stack PDA</i>	True	False
$a^x b^y c^y d^x$ where $x, y \in \mathbb{N}$ is a <i>context-free language</i>	True	False

2. [2 points] Draw a working *PDA* for the following language: $C = \{a^m b^n \mid a, b \in \mathbb{N} \wedge m > n\}$

3. [2 points] Write out a *context-free grammar* for the following language: $D = 1^*0^n1^n0^*$ (Note that this is the language from one of the True/False questions on the last exam)

Module 4: Turing Machines**Name** _____

4. [6 points] Answer the following True/False questions.

A <i>Turing Machine</i> can never run out of space on the <i>tape</i>	True	False
If a language A is <i>decidable</i> , then the complement $\bar{A}$ is also <i>decidable</i>	True	False
<i>Turing Machines</i> can be run (simulated) on other <i>Turing Machines</i> to see how they behave on various input strings	True	False
$a^n b^n c^n$ is <i>NOT context-free</i> , but it is <i>turing-decidable</i>	True	False
Some <i>recognizers</i> never <i>halt and reject</i> on any input	True	False
The <i>turing-recognizable</i> languages are closed under <i>complement</i>	True	False

5. [2 points] Suppose you have a *decider* for the problem/language A (call that machine D). Describe an algorithm that uses D in order to produce a *recognizer* for the complement language $\bar{A}$.6. [2 points] In class we argued that *non-deterministic turing machines* are equivalent in computational power to *deterministic turing machines*. We used three tapes to make this argument. briefly, what was the purpose of each of the three tapes? We don't need a lot of detail here, maybe 1 sentence per tape.

More Context-Free Language Stuff

7. [3 points] For this question, you will show you understand the *pumping lemma for context-free languages* by using it on a specific string. Consider the following language $L = \{a^{3i}b^i \mid \Sigma = \{a, b\}\}$. First, choose a string in this language that has at least 5 characters. Then, show how you will divide this string up into portions to prepare the string to be pumped. Then, pump the string up to $i = 2$ to show how pumping works on this string.
8. [3 points] Suppose I have a *grammar* for a language A with a start variable S_A and another grammar for a language B with a start variable S_B . Describe how I can combine these grammars by adding one new variable and one new rule to create the grammar for $C = A \cup B$ (that's A union B). *HINT: your one new variable will be the start variable of the new union language.*
9. [3 points] Show that $a^n b^n c^n$ is not context-free. You can use any technique you would like.